Driverless Car legislation
Nevada is the first state in the US to have passed a driverless car legislation which means that autonomous cars can be driven on the roads without any restrictions from the Department of Transport

Zero Emission Car?
Nissan's ESFLOW concept car is a 2-seater EV which claims to be a performance oriented sports car that does not give out any emissions

If Cars Could Talk...

Don't start imagining automobiles having conversations while their tanks are being fuelled at the petrol station. But talking cars will be a reality in the near future if research in this field passes the real-life scenario tests. Read on to know more.

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India ranks No.1 in terms of road fatalities, a tag we should surely not be very proud of. According to a World Health Organisation report, the number of accidents per 1,000 vehicles in India is as high as 35 compared to about 4-10 in most developed countries. Even China, which has more vehicles and a higher population has fewer road accidents in comparison.

The reasons for this sorry state of affairs ranges from bad roads to a complete disregard for traffic rules by drivers who are not trained enough to take on the roads. And we're not even talking about drunk driving.

Cars are getting more feature-rich by the day. Safety measures such as air bags, collapsible car material and anti-lock braking systems, among other things are being implemented to reduce accidents. But these features still require the car to be actively controlled by the driver. These technologies are not 'smart' so to speak.

What if your car could interact with other cars on the road? What if your car could anticipate the behaviour of other cars? How about parking your car with your iPod? These might seem like scenarios straight out of a sci-fi novel, but researchers across the world are working on these technologies which are allowing cars to communicate with their surroundings.

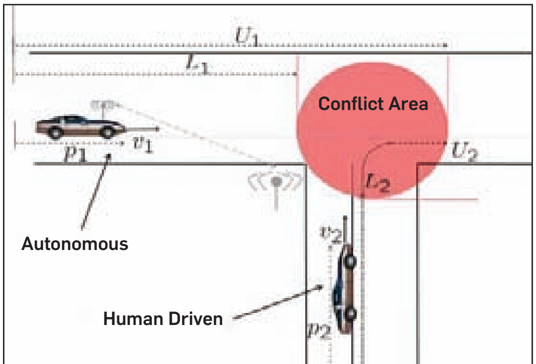
Intelligent collision avoidance system

There have been many developments in accident avoidance systems in cars such as automated cruise control, radar or laser-based sensors, blind-spot warning systems, etc. But these systems are within a car and you still have no control over surrounding cars and how they're being driven, which doesn't make accident avoidance entirely error-free. Mechanical engineers at MIT are working on an intelligent transportation system (ITS) algorithm that takes into account human driving behaviour to warn drivers of potential collisions and takes control of the vehicle to prevent a crash.

The team, comprising Rajeev Varma and Domitilla Del Vecchio, broke down the driving actions of humans to two basic modes: acceleration and braking. Based on the mode the driver is in, there's a finite set of places that the car could be in the next instant or several seconds later. These possibilities, in addition to predictive models of human behaviour, which determine when a driver will slow down or accelerate around an intersection, have been used to make the new algorithm which is being used by the ITS. This algorithm has given rise to a program that can calculate a capture set – a defined area in which two vehicles have a high possibility of a collision. The vehicle which uses this ITS algorithm uses its on-board sensors as well as sensors such as traffic lights to try and predict what the other car (being driven by a human and having no ITS on board) will do and will react accordingly. If both cars are ITS-enabled, then the process becomes quite simple as both the cars can communicate with each other.

The experiment conducted by the team involved two miniature vehicles on two overlapping circles. One vehicle was controlled by a human and the other was autonomous. The human-driven vehicle was controlled by eight different volunteers to simulate different driving behaviours. Out of 100 trials, there were 97 instances of collision avoidance. The vehicles entered the capture set and collided once, thanks to a communication delay between the ITS-enabled vehicles and workstations representing roadside infrastructure that captures and transmits information from non-ITS equipped cars.

In a lab environment it sounds rosy, but



MIT ITS Algorithm